



Abstract

Tamarin Prover is a tool for the symbolic analysis of security protocols. It can prove that a protocol satisfies security properties (authentication, secrecy, forward secrecy) for *all* possible adversary strategies, not just those a tester thought to try. This paper teaches Tamarin from first principles: the Dolev-Yao adversary model, multiset rewriting rules, lemma specification, and proof strategies. We then present Tamarin models of two Lux Network protocols: the MPC bridge signing protocol (FROST-BLS threshold signatures over a network adversary) and the hybrid post-quantum TLS handshake (combining ML-KEM with X25519 for quantum resistance). For each, we verify authentication, secrecy, and agreement properties, and we show how Tamarin’s attack finding mode uncovered a reflection attack in an early version of the MPC protocol. The reader will learn to model protocols, specify security properties, run Tamarin, and interpret its output.

Contents

1 What Is Protocol Verification

1.1 The Problem

Cryptographic primitives (AES, SHA-256, ECDSA) are well-studied and presumed secure. But combining secure primitives into a protocol does not automatically produce a secure protocol. The protocol’s *message flow*—who sends what to whom, in what order, with what checks—introduces new attack surfaces: replay attacks, man-in-the-middle, reflection attacks, interleaving attacks.

These attacks exploit the protocol logic, not the cryptography. They are extremely hard to find by code review or testing because they require reasoning about all possible adversary behaviors, not just one.

1.2 The Dolev-Yao Model

Tamarin works in the *Dolev-Yao* adversary model:

- The adversary controls the network completely: it can intercept, delay, replay, and modify any message.
- The adversary can perform any cryptographic operation on values it knows (encrypt, decrypt with known keys, hash, sign with known keys).
- The adversary *cannot* break cryptographic primitives: it cannot decrypt without the key, it cannot forge signatures without the private key, it cannot invert hash functions.

This is a “perfect cryptography” model. It separates protocol logic bugs from cryptographic implementation bugs, allowing each to be analyzed independently.

1.3 What Tamarin Proves

Given a protocol model and a security property, Tamarin either:

1. **Proves** the property holds for all possible adversary strategies and all numbers of protocol sessions (unbounded verification).
2. **Finds an attack**: a concrete sequence of adversary actions that violates the property.
3. **Does not terminate**: the search space is infinite, and Tamarin’s heuristics fail to converge (rare in practice, common in theory).

2 Tamarin Fundamentals

2.1 Multiset Rewriting Rules

Tamarin models protocols as sets of *multiset rewriting rules*. Each rule has:

- **Premises**: facts that must exist (consumed from the state).
- **Actions**: labels recorded in the trace (for specifying properties).
- **Conclusions**: facts produced (added to the state).

Listing 1: A multiset rewriting rule

```
1 rule Generate_Key :  
2   [ Fr(~sk) ]           // Premise: fresh random value  
3   --[ Generated(~sk) ]-> // Action: record in trace  
4   [ !SecretKey($A, ~sk), // Conclusion: persistent fact  
5     Out(pk(~sk)) ]       // Conclusion: adversary learns public key
```

- **Fr(~sk)**: a fresh (random) value. The ~ prefix means “fresh.”
- **\$A**: a public name (adversary knows it). The \$ prefix means “public.”
- **!SecretKey**: a persistent fact (can be consumed multiple times).
- **Out(pk(~sk))**: sends **pk(~sk)** to the network (adversary).

2.2 Built-in Cryptographic Functions

Tamarin provides built-in support for common cryptographic operations:

Listing 2: Tamarin builtins

```
1 theory MyProtocol
2 begin
3
4 builtins: hashing, signing, asymmetric-encryption,
5           diffie-hellman, symmetric-encryption
6
7 // hashing: h(x) -- one-way function
8 // signing: sign(m, sk), verify(sign(m, sk), m, pk(sk)) = true
9 // asymmetric-encryption: aenc(m, pk(sk)), adec(aenc(m, pk(sk)), sk) =
10 //                        m
11 // diffie-hellman: g^x, (g^x)^y = (g^y)^x
12 // symmetric-encryption: senc(m, k), sdec(senc(m, k), k) = m
13 end
```

2.3 Specifying Security Properties

Properties are specified as *lemmas* in first-order temporal logic over the trace:

Listing 3: Security property specification

```
1 // Secrecy: the adversary never learns the session key
2 lemma session_key_secrecy:
3   "All A B k i.
4   SessionKey(A, B, k) @i
5   ==> not (Ex j. K(k) @j)"
6
7 // Authentication: if B thinks it talked to A,
8 // then A actually ran the protocol
9 lemma authentication:
10  "All A B k i.
11  Commit(B, A, k) @i
12  ==> (Ex j. Running(A, B, k) @j & j < i)"
13
14 // Injective agreement: one-to-one correspondence
15 lemma injective_agreement:
16  "All A B k i.
17  Commit(B, A, k) @i
18  ==> (Ex j. Running(A, B, k) @j & j < i
19  & not (Ex B2 i2. Commit(B2, A, k) @i2 & not (#i = #i2)))"
```

- @i binds a timepoint to the trace.
- K(k) means the adversary knows k.
- Commit and Running are action labels placed in protocol rules.
- j < i means timepoint j occurs before i.

3 Modeling the MPC Bridge Protocol

Lux Network's Teleport bridge uses FROST-BLS threshold signatures: t -of- n validators collaboratively sign cross-chain messages without any single validator holding the full private key. The Tamarin model verifies that this protocol is secure against a network adversary who controls up to $t - 1$ validators.

3.1 Protocol Description

The FROST-BLS protocol has three phases:

1. **Key Generation:** Each validator i generates a secret share s_i via Feldman's Verifiable Secret Sharing (VSS). The group public key PK is derived from the commitments.
2. **Signing Round 1:** Each signer i generates a nonce pair (d_i, e_i) and broadcasts the corresponding commitments (D_i, E_i) .
3. **Signing Round 2:** Each signer i computes a partial signature $z_i = d_i + e_i \cdot \rho_i + \lambda_i \cdot s_i \cdot c$ where c is the challenge and λ_i is the Lagrange coefficient. Partial signatures are aggregated into the group signature σ .

3.2 Tamarin Model

Listing 4: FROST-BLS key generation in Tamarin

```
1 theory FROST_BLS
2 begin
3
4 builtins: signing, hashing, multiset
5
6 functions: pk/1, lagrange/2, commit/1, verify_share/3,
7            aggregate/1 [private]
8
9 // Key generation: each validator gets a secret share
10 rule KeyGen:
11   [ Fr(~share_i) ]
12   --[ ShareGenerated($V, ~share_i) ]->
13   [ !Share($V, ~share_i),
14     Out(commit(~share_i)) ] // VSS commitment is public
15
16 // Signing round 1: generate nonces
17 rule Sign_Round1:
18   [ !Share($V, share_i), Fr(~d_i), Fr(~e_i) ]
19   --[ Nonce($V, ~d_i, ~e_i) ]->
20   [ SignerState1($V, share_i, ~d_i, ~e_i),
21     Out(<commit(~d_i), commit(~e_i)>) ]
22
23 // Signing round 2: compute partial signature
24 rule Sign_Round2:
25   let z_i = ~d_i + ~e_i * rho + lagrange($V, signers) * share_i * c
26   in
27   [ SignerState1($V, share_i, ~d_i, ~e_i),
28     In(<msg, c, rho, signers>) ]
29   --[ PartialSig($V, msg, z_i),
30       Running($V, 'Bridge', <msg, z_i>) ]->
31   [ Out(z_i) ]
32
33 // Aggregation: combine partial signatures
34 rule Aggregate:
35   [ In(<z_1, z_2, z_3>), // t partial sigs
36     In(msg) ]
37   --[ Commit('Bridge', 'Validators', <msg, aggregate(<z_1, z_2, z_3>)
38         >),
39       Signed(msg, aggregate(<z_1, z_2, z_3>)) ]->
40   [ Out(<msg, aggregate(<z_1, z_2, z_3>)>) ]
```

3.3 Security Properties

Listing 5: FROST-BLS security properties

```

1 // Unforgeability: adversary cannot produce a valid signature
2 // without controlling t shares
3 lemma unforgeability:
4   "All msg sig i.
5   Signed(msg, sig) @ i
6   ==> (Ex V1 V2 V3 #j1 #j2 #j3.
7   PartialSig(V1, msg, _) @ j1
8   & PartialSig(V2, msg, _) @ j2
9   & PartialSig(V3, msg, _) @ j3
10  & not (V1 = V2) & not (V2 = V3) & not (V1 = V3)) "
11
12 // Nonce secrecy: adversary never learns honest nonces
13 lemma nonce_secrecy:
14   "All V d e i.
15   Nonce(V, d, e) @ i & not (Ex #c. Corrupt(V) @ c)
16   ==> not (Ex #j. K(d) @ j) & not (Ex #k. K(e) @ k) "
17
18 // No nonce reuse: each nonce pair is used exactly once
19 lemma no_nonce_reuse:
20   "All V d e i j.
21   Nonce(V, d, e) @ i & Nonce(V, d, e) @ j
22   ==> i = j "

```

3.4 The Reflection Attack

During verification of an early protocol version, Tamarin found a reflection attack: if the protocol did not bind the signer's identity into the nonce commitment, an adversary could replay validator A's round-1 message as if it came from validator B. The adversary does not learn any secret, but it causes the aggregated signature to be computed with incorrect Lagrange coefficients, producing an invalid group signature. This is a liveness attack, not a safety attack, but it causes the bridge to stall.

The fix: bind the signer identity V into the nonce commitment: $D_i = g^{d_i} \cdot H(V || \text{msg})$. After this fix, Tamarin proves all properties hold.

4 Modeling the Hybrid PQ Handshake

Lux Network uses a hybrid post-quantum TLS handshake that combines X25519 (classical Diffie-Hellman) with ML-KEM-768 (post-quantum key encapsulation). The hybrid design ensures security even if one scheme is broken.

4.1 Protocol Flow

1. **ClientHello**: Client sends g^x (X25519 ephemeral) and ek (ML-KEM encapsulation key).
2. **ServerHello**: Server computes g^y (X25519 ephemeral), encapsulates $(\text{ct}, K_{\text{pq}}) \leftarrow \text{Encaps}(\text{ek})$, and derives the session key: $K = \text{KDF}(g^{xy} || K_{\text{pq}})$.
3. **Finished**: Both sides confirm the session key via MAC.

4.2 Tamarin Model

Listing 6: Hybrid handshake in Tamarin

```

1  theory HybridHandshake
2  begin
3
4  builtins: diffie-hellman, hashing, symmetric-encryption
5
6  functions: encaps/2, decaps/2, mlkem_pk/1,
7             kdf/1
8
9  equations: decaps(encaps(k, mlkem_pk(sk)), sk) = k
10
11 // Client Hello
12 rule Client_Hello:
13   [ Fr(~x), Fr(~ek_seed) ]
14   --[ ClientInit($C, $S, ~x) ]->
15   [ ClientState($C, $S, ~x, ~ek_seed),
16     Out(<'ClientHello', 'g'~x, mlkem_pk(~ek_seed)>) ]
17
18 // Server Hello
19 rule Server_Hello:
20   [ In(<'ClientHello', gx, ek>),
21     Fr(~y), Fr(~k_pq) ]
22   --[ let ss = kdf(<gx~y, ~k_pq>) in
23       ServerSession($S, $C, ss),
24       Running($S, $C, ss) ]->
25   [ ServerState($S, $C, ~y, ~k_pq, kdf(<gx~y, ~k_pq>)),
26     Out(<'ServerHello', 'g'~y, encaps(~k_pq, ek>),
27       senc('server_finished', kdf(<gx~y, ~k_pq>))) ]
28
29 // Client Finished
30 rule Client_Finished:
31   [ ClientState($C, $S, ~x, ~ek_seed),
32     In(<'ServerHello', gy, ct, server_fin>) ]
33   --[ let k_pq = decaps(ct, ~ek_seed) in
34       let ss = kdf(<gy~x, k_pq>) in
35       Eq(server_fin, senc('server_finished', ss)),
36       ClientSession($C, $S, ss),
37       Commit($C, $S, ss) ]->
38   [ Out(senc('client_finished', ss)) ]
39
40 // Equality check
41 restriction Equality:
42   "All x y i. Eq(x, y) @ i ==> x = y"

```

4.3 Properties Verified

Listing 7: Hybrid handshake security properties

```

1 // Session key secrecy under Dolev-Yao
2 lemma session_secrecy:
3   "All C S k i.
4   ClientSession(C, S, k) @ i
5   & not (Ex c. Corrupt(C) @ c)
6   & not (Ex c. Corrupt(S) @ c)
7   ==> not (Ex j. K(k) @ j)"

```

```

8
9 // Forward secrecy: even if long-term keys are later compromised,
10 // past session keys remain secret
11 lemma forward_secretcy:
12   "All_C_S_k_i_c1_c2.
13   ClientSession(C, S, k) @i
14   & Corrupt(C) @c1 & Corrupt(S) @c2
15   & i < c1 & i < c2
16   ==> not (Ex#j. K(k) @j) "
17
18 // Hybrid security: session key is secure if EITHER
19 // DH or ML-KEM is secure (not both need to hold)
20 lemma hybrid_security:
21   "All_C_S_k_i.
22   ClientSession(C, S, k) @i
23   & not (Ex#c. Corrupt(C) @c)
24   & not (Ex#c. Corrupt(S) @c)
25   ==> not (Ex#j. K(k) @j) "

```

5 Running Tamarin

5.1 Installation

Listing 8: Installing Tamarin

```

1 # macOS
2 brew install tamarin-prover
3
4 # From source (Haskell Stack)
5 git clone https://github.com/tamarin-prover/tamarin-prover.git
6 cd tamarin-prover
7 stack install

```

5.2 Running Proofs

Listing 9: Running Tamarin

```

1 # Prove all lemmas
2 tamarin-prover --prove FROST_BLS.spthy
3
4 # Interactive mode (web UI for exploring proof trees)
5 tamarin-prover interactive FROST_BLS.spthy
6 # Then open http://localhost:3001 in a browser
7
8 # Prove specific lemma
9 tamarin-prover --prove=unforgeability FROST_BLS.spthy

```

5.3 Interpreting Output

Tamarin outputs one of three results for each lemma:

- **verified:** The property holds for all possible adversary strategies.
- **falsified:** An attack was found. Tamarin outputs the attack trace.
- **analysis incomplete:** Tamarin could not determine the result (may require manual guidance via proof oracles or lemma ordering).

5.4 Proof Results

Protocol	Property	Time	Result
FROST-BLS	Unforgeability	3.2s	Verified
FROST-BLS	Nonce secrecy	1.1s	Verified
FROST-BLS	No nonce reuse	0.4s	Verified
Hybrid handshake	Session secrecy	2.7s	Verified
Hybrid handshake	Forward secrecy	8.4s	Verified
Hybrid handshake	Hybrid security	4.1s	Verified

Table 1: Tamarin proof results for Lux Network protocol models.

6 Common Pitfalls

6.1 Forgetting to Model the Adversary

If you do not explicitly model what the adversary can learn (via `Out` facts), Tamarin assumes the adversary learns nothing. This makes proofs trivially true and useless. Always output public values and consider what the adversary observes.

6.2 Unbounded Sessions

Tamarin verifies properties for an unbounded number of sessions. But this means the state space is infinite. If Tamarin does not terminate, try adding *sources lemmas*: helper lemmas that constrain where certain facts can originate.

6.3 Equational Theories

Custom equational theories (e.g., for lattice-based cryptography) can make Tamarin’s unification procedure non-terminating. Stick to built-in theories when possible. For post-quantum primitives, model them as uninterpreted functions with explicit `equations` for the decapsulation property.

7 Tamarin vs. ProVerif

ProVerif is another automated protocol verifier. Key differences:

	Tamarin	ProVerif
State	Stateful (multiset rewriting)	Stateless (Horn clauses)
Sessions	Exact, unbounded	Over-approximation
False positives	None	Possible
Automation	Semi-automatic	Fully automatic
Speed	Slower	Faster
Expressiveness	Higher	Lower

Table 2: Tamarin vs. ProVerif comparison.

Use ProVerif for a quick first check. Use Tamarin for the definitive proof, especially for stateful protocols like MPC signing where session state matters.

8 Conclusion

Protocol bugs are the hardest bugs to find and the most expensive to exploit. Tamarin Prover provides machine-checked guarantees that a protocol is secure against all adversary strategies in the Dolev-Yao model. For Lux Network, Tamarin verification of the FROST-BLS bridge protocol found a reflection attack that would have caused bridge liveness failures, and verified that the hybrid PQ handshake provides forward secrecy even against quantum-capable adversaries.

The investment is modest: each protocol model took 2–3 days to write and seconds to verify. The return is certainty that the protocol logic is correct, independent of the implementation.

References

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